

Predictive deep learning explains human BOLD responses during natural viewing

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MAX-PLANCK-GESELLSCHAFT

Background

- Hypothesis: the brain (somehow) implements predictive coding [1,2,3]
 - Minimizing prediction error requires no labels
 - Typically tested with artificial stimuli
 - Difficult to derive predictions
- Deep Neural Networks (DNN) can model fMRI activity related to object recognition [4, 5]
- But how well can a **predictive coding** DNN [6] model fMRI activity under **natural viewing conditions**?

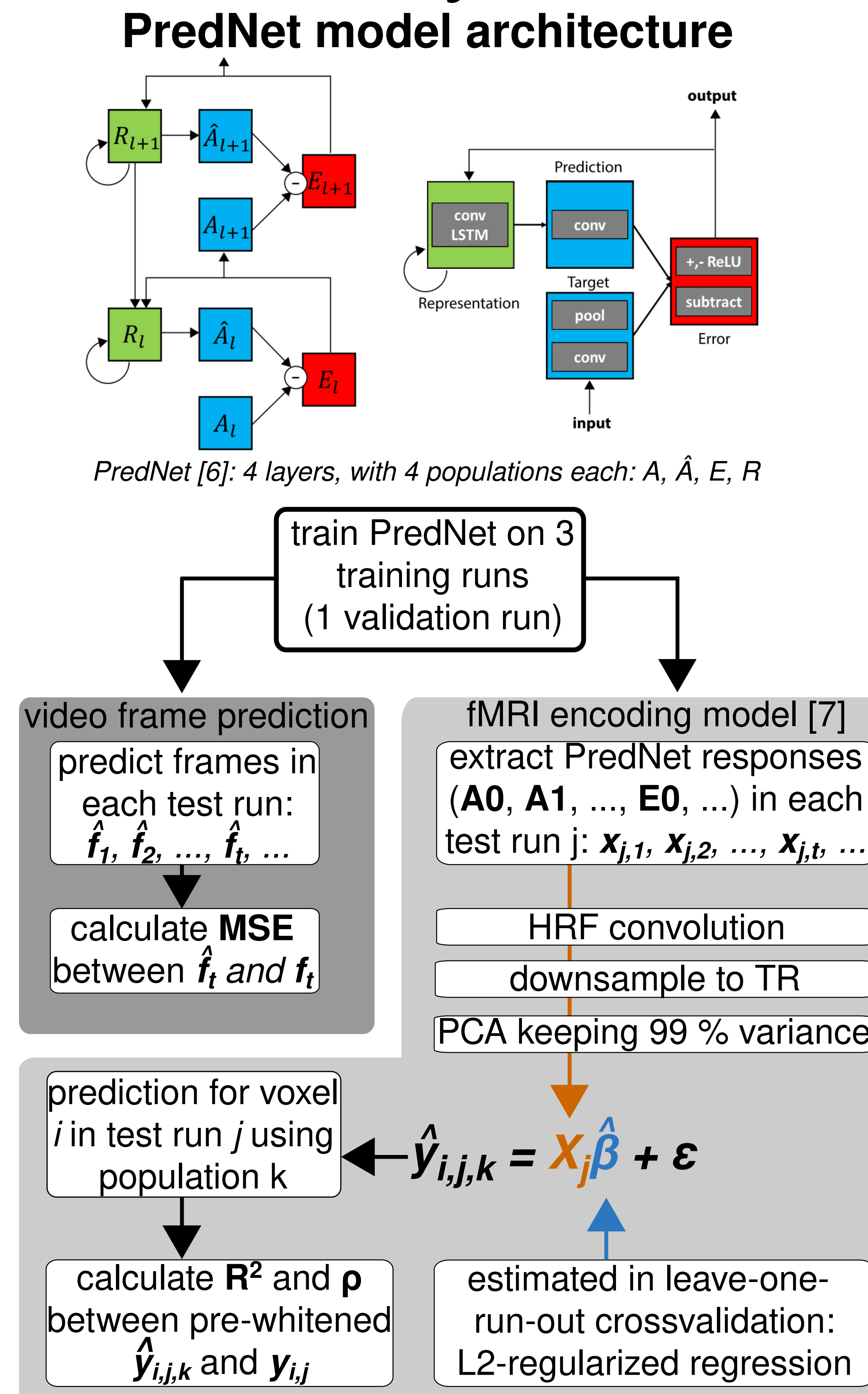
Experimental design



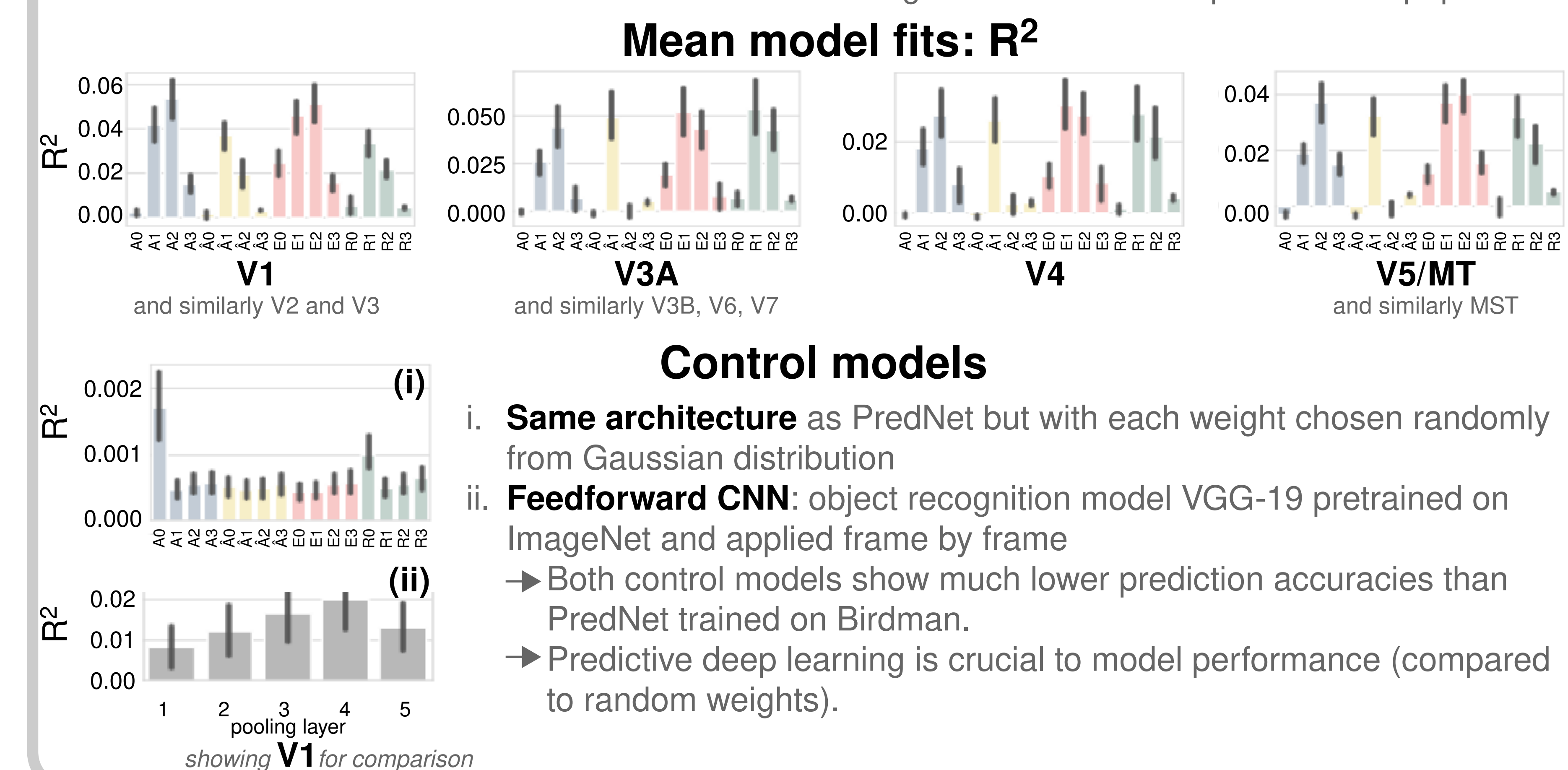
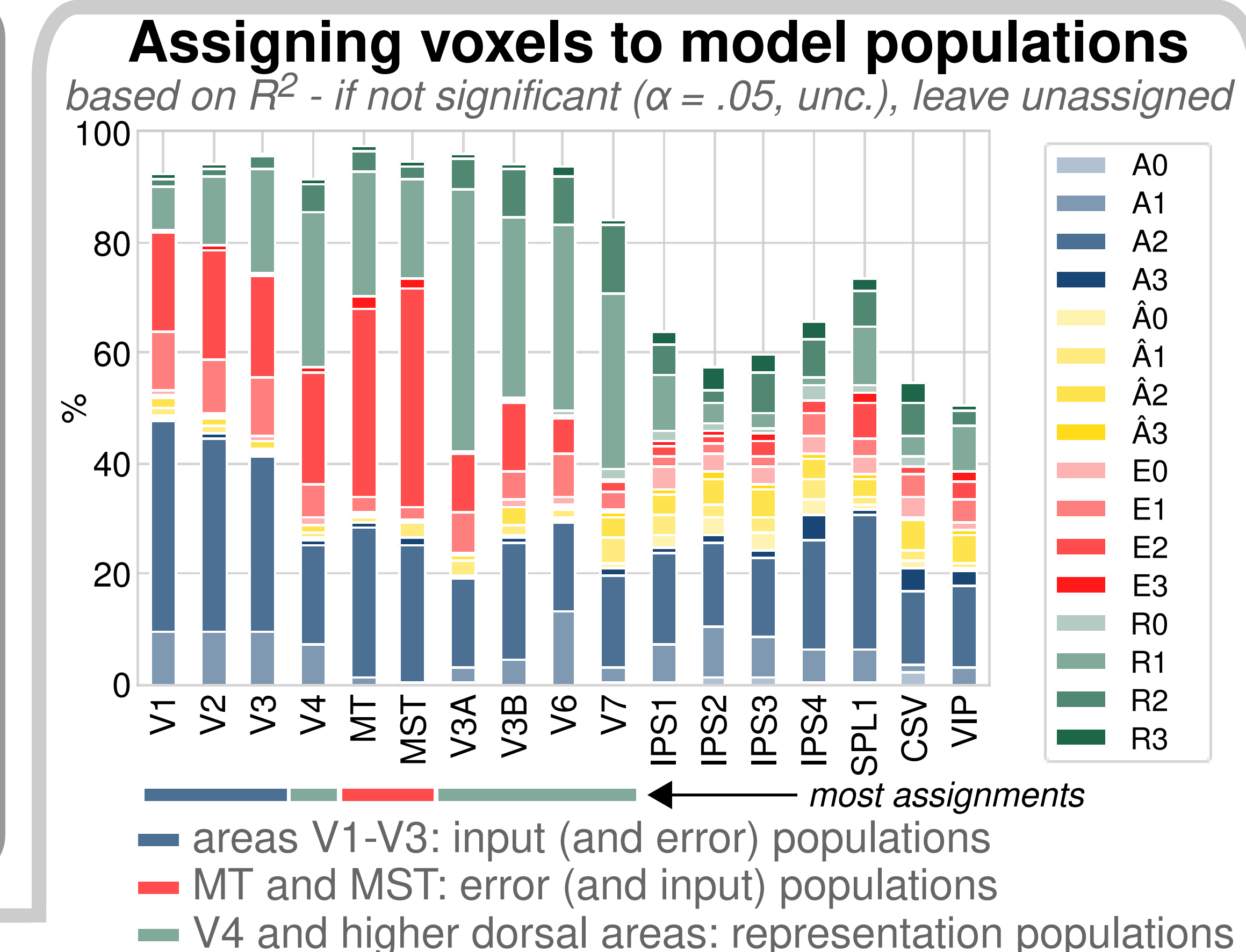
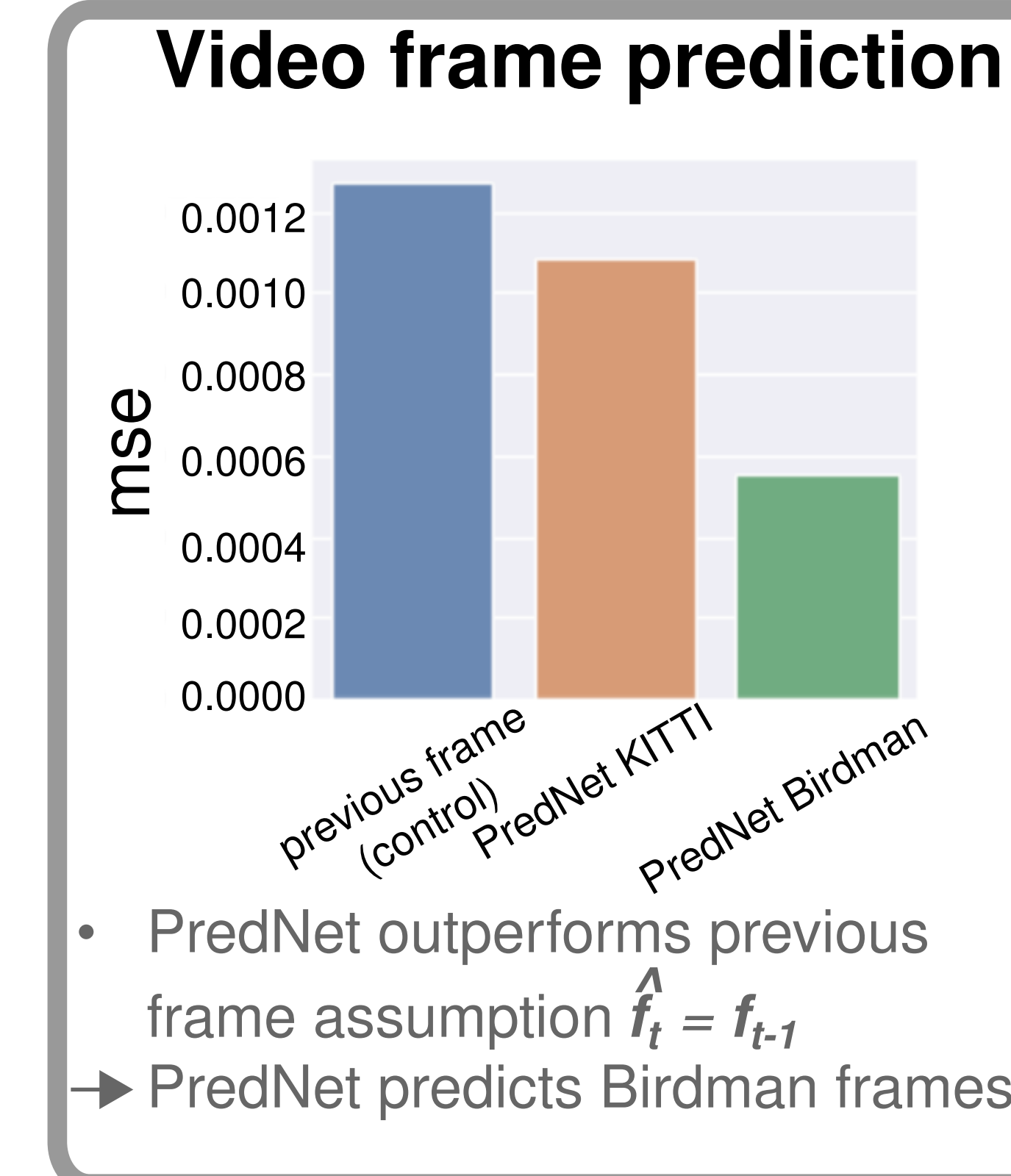
Birdman (Fox Searchlight Pictures, 2014)

- 12 participants watched 50 minutes of the movie “Birdman” in the MRI scanner in 5 runs (10 minutes each)
- Free viewing instruction
- ROIs identified individually using retinotopic mapping and motion localizers
- fMRI: $B_0 = 3$ T, $TR = .87$ s, $TE = 30$ ms, 57° flip angle, $2 \times 2 \times 2$ mm³, 56 slices

Data analysis



Results



Conclusion

- PredNet can predict the next video frame in feature movies while explaining human BOLD activity measured concurrently.
- EVC activity explained best by input (and error) populations
- MT+ activity explained best by error (and input) populations
- V4 and higher dorsal areas explained best by representation populations

References

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